



Physical laws meet machine intelligence: current developments and future directions

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Abstract

The advent of technology including big data has allowed machine learning technology to strengthen its place in solving different science and engineering complex problems. Conventional deep machine learning algorithms work as a black box while dealing with various complex physics-driven problems. This problem can be reduced by integrating the physical laws with machine learning algorithms to ensure the developed models are complied with the physics and are potentially more explainable. This physics-informed machine learning (PIML) approach allows the integration of physical laws in the form of PDEs into the loss function of the neural network, hence, constraining the training of the complex problems based on both the physical, experimental, and mathematical boundaries. This, hence, allows the development of a more general predictive model for different science, engineering, and optimization tasks. Considering such advancements in the machine learning domain, this review presents the systematic progress in the development of integrating physics into the neural networks and recent applications in solving various forward and inverse problems in science and engineering. This paper can serve as a reference for the researchers, developers, and users to get all information they need before developing, implementing, and deploying AI models and smart systems that are equipped with the PIML methodology. It highlights the benefits and points out its limitations and recommendations for further development. The review also compares the traditional data-driven machine learning and PIML approach in dealing with the physics of complex problems. In general, the PIML has been found to provide consistent results with the exact solutions and physical nature of the system. However, similar to other AI system development, a more robust and complex AI algorithm requires more computational power which is also the case in PIML development and implementation. It should be noted that different terminologies such as physics-informed neural networks (PINN), science-informed neural networks, physics-inspired neural networks, and physics-constrained neural networks have been used in the literature that describes the very similar concept of integrating physical laws with machine intelligence. For consistency, we use the PIML term throughout this paper which covers all listed terminologies in this regard.

Keywords Physics informed machine learning · Physics informed neural network · Loss function · PDEs · Explainability · Physics compliance

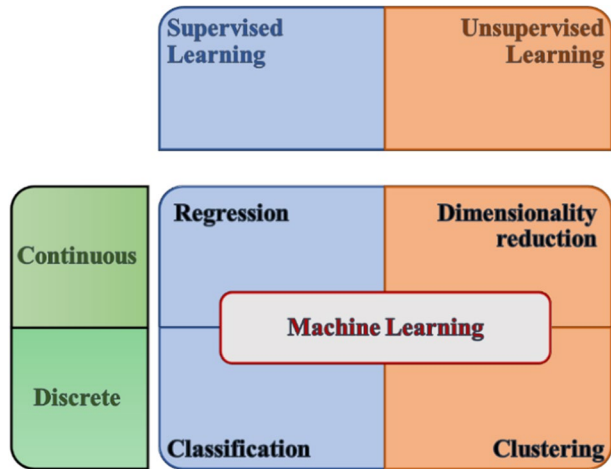
1 Introduction

The development of technology over the past decade has changed the global scenario of solving science and engineering problems. Different modeling and prediction techniques have taken a huge role in solving underlying problems. Even with such improvements, yet, modeling and simulation always remain problematic because of having complicated nature of the system. Although, different mathematical and numerical techniques, especially involving partial differential equations (PDEs), are developed, yet, they are based on assumptions to solve the complex nature of problems. This assumption, however, possesses huge uncertainty due to the lack of a large number of features controlling the system and physics involved. Different mathematical techniques containing limited physical features while solving numerical models requires huge resources which extend from complex formulations to expensive computational solvers.

To understand the nature of the system, without relying on mathematical and numerical simulation, observational data plays its part. Due to technological advancements, a lot of data has been obtained through various sources such as sensors, testing, and physical observations. This has allowed the analysis of the various science and engineering processes with ease (Reichstein et al. 2019). However, it also has some restrictions. The major restriction involves the analysis of the large dimensional data which is very hard to evaluate. They also restrict generating the trend analysis and visualizing the impacts of various parameters on a single platform. Again, the integration of this multi-dimensional data with the existing mathematical and numerical correlations is difficult.

To cope with this restriction and the availability of large computational data, machine learning (ML) has expanded the space of the analysis. It is one of the emerging and fastest-growing technology that has replaced human intelligence with machines. Through different machine learning approaches, one can easily generate various correlations, design various systems, optimize the nature of the analysis, and understand the process descriptions. For instance, one ML algorithm is used to predict variations in CO₂ fluxes (Papale and Valentini 2003). In other examples, it is used to predict soil types, detect an anomaly in different structures, or space-time-dependent state predictions (Grimm et al. 2008; Hengl et al. 2017; Zhang et al. 2016; Ball et al. 2017). These algorithms can work both in a supervised and unsupervised way with most of the models based on supervised learning (refer to Fig. 1) (Fahle et al. 2020). The supervised methods work on generating a model based on the training of both the input and output data. Also, they work on both the regression and classification problems. The supervised algorithms involve iteratively training the data input and the output data and optimizing it until reaching the minimal difference between the actual value and the predicted value. Most common supervised learning algorithms include Artificial neural networks (ANN), support vector machines (SVM), decision trees, random forest (RF), k-Nearest Neighbor (kNN) algorithms, linear and logistic regression, etc. Unsupervised learning is based on grouping and interpreting the input data only. They automatically determine the dataset organization while best describing the dataset based on its input features. Generally, this learning is categorized in the family of clustering analysis, however, dimensionality reduction has also its roots in it. The clusters are generally deduced using the probabilistic, similarity, or Euclidian distance measures. Commonly used clustering algorithms include k-Means clustering, hierarchical clustering, hidden Markov models, self-organizing maps (SOM), and Gaussian mixture models (Ippolito et al. 2021).

Fig. 1 Machine learning approaches and characterization



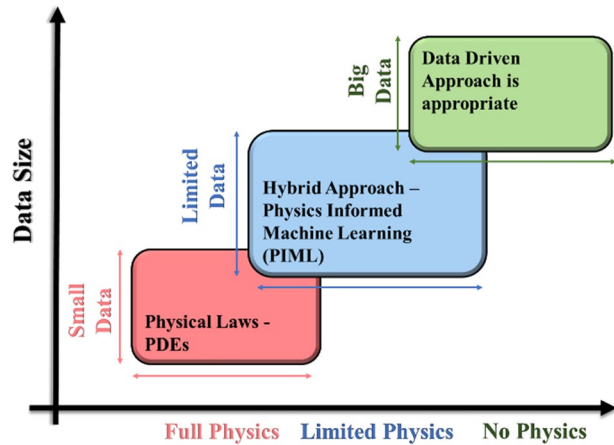
Again, fitting the machine learning models with a large amount of data (which is conventionally being done) provides some limitations. Although they accurately predict the data, their structures, and impacts, however, their extrapolation many times becomes misleading. The algorithm alone predicts data accurately within the range it is trained but it gets difficult to explain the applicability of the algorithm outside the range. Hence, interpretability and physical consistency is a concern. An added disadvantage is the presence of uncertain data or outliers which may mislead the whole training predictability (Montavon et al. 2018; Runge et al. 2015; Ghahramani 2015; Goodfellow et al. 2014). This is where it becomes important to train the current algorithms about the physical laws governing those trends. Hence, the need for new optimized algorithms i.e. physics informed machine learning (PIML) algorithms becomes highly important which constrains the training of the problems based on physical and mathematical boundaries. This, hence, allows the development of a more general predictive model for different engineering and optimization tasks.

Considering such advances in machine learning and the need for physics-informed architecture, this review paper aims to understand the methodology behind different physics-informed machine learning approaches. The paper also presents different applications that are implemented in sciences and engineering as part of the PIML family. Different databases and libraries are presented as part of the review. At last, conclusions are drawn and comments are provided concerning the applicability, advantages, and shortfalls of this technique.

2 Data and physics

For machine learning problems, it is accepted that a large amount of data is required for accurate modeling of the system. To define the physics as well, a large dataset is always required so the algorithms can understand various variations in the system and train the model accordingly. Karniadakis et al. (2021) divided the data-physics paradigm into three regions of complexity. In the first region (refer to Fig. 2), there is very small data available including the initial and boundary conditions and PDE based on different physical laws, and correlations are solved to quantify the relationships between the variables. This is the

Fig. 2 Data–physics paradigm
Modified from Karniadakis et al.
(2021)



case of many simulators in use today where the inputs are generally defined, the simulator solves PDE and provides different relationships based on the physical correlation. In the last region, a lot of data is available, as discussed earlier, due to the availability of technological resources today. Hence, training ML models on such datasets and devising the physical relationship within the domain of trained data is highly possible. This approach is very much suitable with high dimensional data, where it is difficult to characterize a system in terms of simple physical equations. This scenario, in general, corresponds to data-driven modeling. The problem is with the middle region where we have limited data and limited physics available. This is the region of interest, where the integration of physics and governing physical equations, as in the case of PIML workflow, plays its part and develops a productive predictive model (Karniadakis et al. 2021).

Considering such data and physics relationships, the physics-informed models can be generalized into two approaches (1) the data-driven approach and (2) physics-informed machine learning. The data-driven approach takes an appropriate amount of data and defines the problem within the range of the training set, while, the PIML approach results in the integration of both physical laws and a limited amount of data. Even with the data-driven approach, a hybrid methodology is possible that works with integrating the big data and physical laws, however, this process will result in high computational time and resources.

3 Data-driven modeling approach

The application of data-driven modeling has increased significantly in the last decade. Its application has become extensive with the advancement in machine learning techniques. With a large amount of data available due to the latest technology implications, data-driven modeling has spent its roots replacing many physics-based models in various disciplines including applied sciences and engineering (Reichstein et al. 2019; Bergen et al. 2019; Hsieh 2009; Karpatne et al. 2017, 2018; Kutz 2017; Wang et al. 2018).

The physical models have a variety of applications in the aforementioned disciplines. Many times, however, they present far from reality approximations because of the complex nature of the processes involved in the systems. This is the result of various constraints

underlying while developing such models. Also, the heterogeneous parameters of the complex systems in these first-principle models are estimated through limited observational data, reducing their performance further. This is where the machine learning models have been found to outperform the physics-based models in their various applications to sciences and engineering (Kauwe et al. 2018; Wei et al. 2018; Ibarra-Berastegi et al. 2015). The reason for such performance is the ability of ML methods to identify multidimensional features, patterns, and structures which is difficult in the case of constrained physical models (Willard et al. 2020).

The data used in this modeling category usually comes through two sources (1) observational data including data from experiments, sensors, and/or physical monitoring (2) generated data through analytical and numerical modeling. This data is usually trained to model complex processes. Often, the data generated through numerical modeling is physics-guided as the data points are generated based on the solutions of complex equations defining the system. Generally in a data-driven approach, the data has been initially pre-processed which includes the determination of relative importance and orthogonality of the features, removal of outliers, and data minimization to prevent overfitting issues. After preprocessing, the data is distributed into training, testing, and validation sets. As a rule of thumb, the data is usually distributed with 80–20% or 75–25% of training and testing data respectively. After this, the model is trained with the training set while minimizing the loss function and then tested to determine the performance of the ML model on unseen data. This is how a data-driven model is developed for results prediction and forecasting. It is important to note that the model should be updated with the latest data so that the model performance remains consistent (refer to Fig. 3) (Liu et al. 2021).

There have been multiple works of literature reported on such data-driven implementation, for instance, Sherman et al. (2019), in their approach use physics informed DNN framework to interpret distributed acoustic sensors (DAS) measurements in real-time (Sherman et al. 2019). In the energy industry, data-driven approaches have been used for determining the well performance in hydrocarbon reservoirs with different degrees of success (Crnkovic-Friis and Erlandson 2015; Syed et al. 2021a, 2022a; Luo et al. 2018; Muthur et al. 2022a; Liang et al. 2019; Liao et al. 2020). In addition, various algorithms including ANN, convolution neural networks (CNN), recurrent neural network (RNN), SVM, Long short-term memory (LSTM), RF, kNN, etc. have been used for exploration, characterization, and production technology (Sprunger et al. 2021; Syed et al. 2020b, 2021b, 2022b, c; Sattarin et al. 2021a, b). Other literature also focuses on generating a physics-driven dataset from numerical simulators and integrating it with ML models for decarbonization to predict CO₂ storage mechanisms. For instance, Thanh et al. (2020) in their study, applied ANN to predict the CO₂ storage capacity in the residual oil zone. They generated 351 numerical datasets based on uncertain parameters including the reservoir properties such as porosity, permeability, and residual saturations. and thickness; and well properties such as bottomhole pressure and injection rate. The generated points contain cumulative CO₂



Fig. 3 Step-by-step process for data-driven machine learning

retained, CO₂ storage, and oil production as an output. Later, they used ANN to train the generated dataset and determined an accurate predictive model with a coefficient of determination (R^2) of over 98%. Further, for the validation of their model, they tested the model with five field cases from the Permian basin. This validation suggested their model results contain a good agreement with the field cases (Thanh et al. 2020). In another similar work, Song et al. (2020) used ANN for understanding the CO₂ sequestration process in a saline aquifer (Song et al. 2020). Similar to this workflow, different works of literature integrated the numerical simulator results with ANN to develop models for predicting the geothermal reservoir and well performances, and pressure and temperature predictions (Zhou et al. 2019; Aydin et al. 2020; Pérez-Zárate et al. 2019; Muther et al. 2022b).

In addition, various pieces of literature presented the application of the data-driven approach in pipeline engineering. Large data coupled with machine learning algorithms were used for pipeline integrity management, real-time leakage detection, corrosion detection, pipeline failure, and surface gas transportation (Li et al. 2021b; Seghier et al. 2020; Rachman et al. 2021; Soomro et al. 2022; Ayati et al. 2022; Mazumder et al. 2021; Gupta and Li 2022). Moreover, different works presented the application of ML algorithms including ANN, SVM, and CNN coupled with different optimization algorithms in civil and geotechnical engineering. The applications include structural health monitoring, design and optimization, vibration control, damage detection, system identification, pile foundations analysis, concrete crack detections, water treatment plant monitoring, bridge design, corrosion behavior, hydropower generation, and so on (Hsieh and Tsai 2020; Xie et al. 2020; Adeli 2001; Amezquita-Sanchez et al. 2016; Shahin 2015, 2016; Cho et al. 2018; Reich 1997; Sapitang et al. 2020).

Such a data-driven approach has been expanded for earthquake detections and warnings (Perol et al. 2018; Kong et al. 2016; Reddy and Nair 2013; Ochoa et al. 2018), categorizing particles of volcanic ashes (Shoji et al. 2018), studying mantle convection processes (Shahnas et al. 2018), noise tomography (Paitz et al. 2018), and geological structure development (Chan and Elsheikh 2017; Bergen et al. 2019). Wiszniowski et al. (2014) used RNN for the development of a workflow for detecting earthquakes in real-time. Similarly, Moseley et al. (2018) performed full fast wavefield simulations while using CNN on synthetically generated finite difference data. The data-driven approach has also been used in ocean, climate, and weather modeling (Adie et al. 2018). In climate and weather modeling, a massive climate and weather dataset is used to predict weather events such as tropical cyclone detection, weather forecasting, precipitation, spatiotemporal features, and downscaling the climate variables (Kim et al. 2017a, b; Mouatadid et al. 2017; Seo et al. 2017; Matsuoka et al. 2017). In ocean engineering, the same approach is used to predict ocean temperature, salinity, and forecast wave conditions such as wave height and wave characteristic period (Bhaskaran et al. 2010; Makarynskyy 2004; Browne et al. 2007; James et al. 2018).

Similar approaches have also been followed in many other disciplines such as in quantum mechanics different ML algorithms including Feed-forward NN (FFNN), SVM, kNN, kernel functions, Restricted Boltzmann Machine (RBM) architecture, and non-linear ridge regression have been used for ground-state calculation of molecular systems instead of solving computationally expensive Schrodinger's equation (Rupp 2015; Rupp et al. 2012; Hansen et al. 2013; Carleo and Troyer 2017). In material science, ANN has been widely used for characterizing the material. Specific applications include the understanding of the stress-strain relationship (Ghaboussi et al. 1990, 1991), material investigation, compressive strength calculations (Yeh 1998), and analyzing the structural dynamics (Zimmerman et al. 2005). The material research has been extended to nanofluids calculations including thermal conductivity and viscosity calculations (Esfe et al. 2015a, b). In addition, different

ML algorithms have been utilized to estimate and optimize the outcomes of chemical reactions for material design (Wang et al. 2020). At the molecular level, the ML algorithms are also implemented to determine potential energy surfaces (PES), for the first attempt by Sumpter and Noid (1992). Later, Blank et al. (1995) presented the first DFT-based ML PES to represent potential energy surfaces. This work uses a traditional (FFNN) to achieve this objective. In addition, different neural network potentials (NNPs) are used for various problems including low-dimensional system construction including reactive molecular systems (Le and Raff 2010; Jiang and Guo 2013; Li et al. 2013), small molecules (Raff et al. 2005; Brown et al. 1996; Manzhos et al. 2006; Prudente and Neto 1998; Rocha Filho et al. 2003), and their interaction with metal surfaces (Blank et al. 1995; Lorenz et al. 2004, 2006; Ludwig and Vlachos 2007; Behler et al. 2008; Carbogno et al. 2008).

In medical science, the data-driven ML approach has been quite popular in medical diagnosis classification, detection of cancer specifically lung cancer through CT scans, MRIs, and X-Rays; disease prognosis and diagnosis; prediction of a disease outbreak; prediction of mortality rate; brain tumor identification; heart disease; and many more (Wang et al. 2019; Dallora et al. 2017; Chen et al. 2017; Chaurasia and Pal 2014; Garg and Mago 2021; Frank et al. 2020; Taylor et al. 2016; Krishnaiah et al. 2013; Kuruvilla and Gunavathi 2014; D'Cruz et al. 2016). Apart from this, studies have also been conducted utilizing data-driven and conventional AI approaches for understanding blood flow (Arzani et al. 2021).

The presented works of literature are some of the examples of large literature available that utilizes the data either generated through numerical simulators or obtained from the actual material/resource to develop a data-driven model for system prediction. The general and well-represented workflow (although some variations are possible) followed by most of the literature is presented in Fig. 4.

Although, as presented, such an approach has been widely successful in many applications, however, it is still a black box with no physical constraints defined in the system. The same data-driven approach, in the literature, has also been introduced by many names such as physics-guided, theory-guided, physics-based, etc. because of using large physical data for training, however, again no physical restrictions are enforced in the training process. This (data-driven) methodology is widely useful in understanding multi-dimensional physical problems with the capability of capturing intricate spatiotemporal relationship that is not fully understood because of the complexity involved in the process. Also, this approach appears to predict very well within the range of its training set. However, it

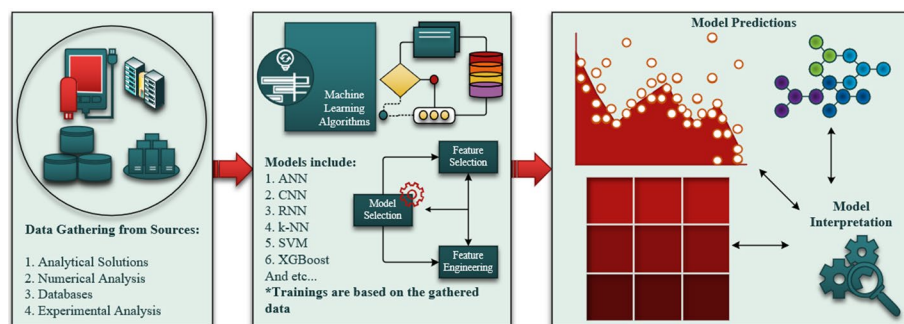


Fig. 4 General approach adopted for data-driven modeling

has some concerning limitations (1) they require a large amount of dataset which is not readily available for all the processes, and large computational power to generate synthetic relationships (2) the reliance on results outside the training region is still questionable, and hence, the generalization of the model may suffer severe inconsistencies. Again, this method, although, is not physics-oriented as the laws are not defined, yet, a large system of dataset aid in guiding in ML model to learn the physical relationships. In any case, with limited data, this approach will present a variety of concerns, hence, the integration of physical laws with the power of ML aid in the accurate development of the physics-guided phenomenon.

4 Physics informed machine learning (PIML) approach

As discussed earlier the limitations of using a data-driven modeling approach with limited data, efforts have been made to develop a more robust ML framework that can incorporate the physical nature of the system. In this effort, the workflows including informing machine learning algorithms, especially in the neural networks, with physics have shown extensive success in approaching the problem with limited data. This technique i.e. physics informed machine learning (PIML) embeds physical laws (including PDEs) with the neural network to achieve a physically consistent model. Initially, the concept of using neural networks to solve ODEs and PDEs have been introduced in the 1990s in various works for initial and boundary values conditions. The main advantage of using neural networks to solve PDEs is its universal approximation capabilities representing any PDE while avoiding discretization steps and overcoming the curse of dimensionality issues (Lagaris et al. 1998, 2000; Lee and Kang 1990; Meade and Fernandez 1994a, b; Milligen et al. 1995; McFall and Mahan 2009; Shirvany et al. 2009; Kumar and Yadav 2011; Mall and Chakraverty 2016; Ryck and Mishra 2021). Recent approaches by Raissi et al., especially their physics-informed neural network (PINN), have resulted in promising results in different PDE problems for the physics-informed machine learning domain (Raissi et al. 2017, 2018, 2019; Raissi and Karniadakis 2018). With the development of PINN, various workflows such as PIDL, MPINN, Physics constrained DNN, PIANN, PhyCRNet, FINO, etc., have been presented by various researchers with the same goal of incorporating physics into the neural network. Specifically, in a physics-informed machine learning setup, different PDEs (such as stochastic, differential, integer-order, integro-differential) along with their respective initial conditions (ICs) and boundary conditions (BCs) are integrated into the loss function of a neural network as (Raissi et al. 2019; Lu et al. 2021a; Pang et al. 2019; Zhang et al. 2020; Dissanayake and Phan-Thien 1994; Cuomo et al. 2022; Shukla et al. 2021b; Cai et al. 2022):

$$Loss_{total} = w_{PDE}L_{PDE} + w_{data}L_{data} + w_{IC/BC}L_{IC/BC}, \quad (1)$$

where L_{data} represents the loss generated through the training dataset (loss from initial and boundary conditions of a PDE), while L_{PDE} represents loss due to PDE which constrains the neural network to physical phenomena in the PDE. The PIML algorithms are trained by minimizing the loss function defined in Eq. 1 through different gradient-boosting optimizers (Kingma and Ba 2014; Byrd et al. 1995; Bengio 2000; Kylasa et al. 2019; Richardson 2018). The optimization is stopped once the threshold value of the loss is reached. In summary, this physics-informed solution involves the distribution of the neural network into two parts (i) the first part takes input features as a traditional-black box ANN with several hidden layers (ii) the second part contains physical equations where the derivatives are used

as an activation function. Both these parts are linked together and contribute to the loss of the overall training process.

This process allows limited data to be incorporated, and hence, it speeds up the training process in terms of data size requirements. This PIML approach can be used to solve both the forward and inverse problems containing non-linear PDEs. The forward problem includes the solutions to PDE providing appropriate initial/boundary conditions while the inverse problem allows the network to obtain and identify PDE parameters from the training dataset (Raissi et al. 2019; Zhang et al. 2020; Tipireddy et al. 2019; Guo et al. 2020). A general framework of the PINN approach based on the above discussion is presented in Fig. 5.

Different optimizers such as Adam, stochastic gradient descent (SGD), root mean square propagation (RMSprop), and Broyden–Fletcher–Goldfarb–Shanno (BFGS) have been used to minimize the highly nonconvex and nonlinear loss functions in PIML. The most popular ones are the Adam optimizer which is the first-order stochastic gradient-based optimizer (Kingma and Ba 2014) and L-BFGS (which is a limited-memory function of BFGS) that uses the Hessian matrix (second-order) for optimization direction calculations (Byrd et al. 1995; Fletcher 2013). When compared with the Adam optimizer, L-BFGS results in accurate results with fewer iterations because of second-order derivative calculations for smoother PDEs, however, it quickly converges to the residual local minimum in more stiff PDEs. Therefore, L-BFGS is sometimes used with an Adam optimizer to avoid local minima for a more refined solution (Lu et al. 2021a; Markidis 2021). In one of the literature, particle swarm optimization (PSO) optimizer (a metaheuristic optimization algorithm)

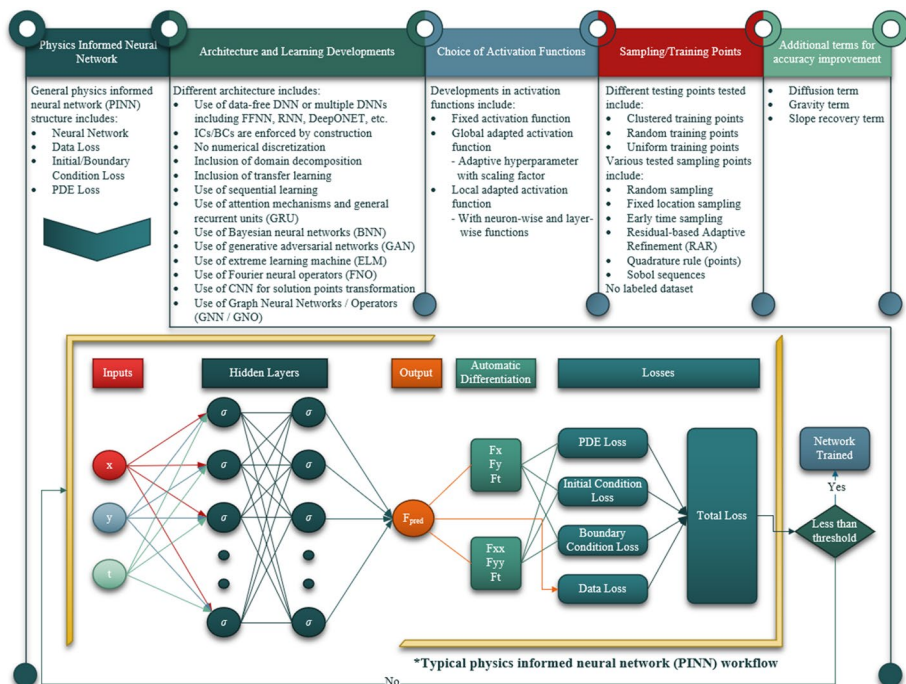


Fig. 5 Major current developments in PIML workflow for predictive modeling

(Kennedy and Eberhart 1995) is also tested and compared with Adam optimizer in solving linear and non-linear PDEs. The results show better optimization through PSO than Adam optimizer (Davi and Braga-Neto 2022).

This PIML architecture has been developed over years with a large number of updates. Various architecture developments such as fully connected neural networks (FC-NN), FFNN (Raissi and Karniadakis 2018), CNN (Gao et al. 2021), DeepONet (Lu et al. 2021b), etc. have been presented to optimize the network performance while constraining the network training through PDEs. To tackle computational expenses different activation functions such as globally and locally adaptive functions, and various architectures such as conservative PINN (cPINN), extended PINN (XPINN), CNN implementation, etc. with different parallelization capabilities have been used. Different parallel computing algorithms have been proposed by various researchers such as the approach by Shukla et al. (2021b) and parareal physics-informed neural network (PPINN) (Meng et al. 2020) that distribute larger tasks into smaller tasks allowing the training of the machine learning algorithms to solve PDEs with comparably less computational time on multiple CPU-GPU cores.

Apart from the integration of physical laws into different neural networks, there are other approaches also presented in PIML frameworks. In work by Raissi and Karniadakis (2018), it can be observed that the Gaussian process can very well infer the hidden physics of a known governing PDE with the availability of noisy measurements. To support the proposed work, they tested the methodology on Burgers' equation, Kuramoto–Sivashinsky equation, The KdV equation, Schrödinger equation, Navier–Stokes equations, and fractional equations. In other literature, Yu (2018) presented the Deep Ritz method (DRM) to solve variational problems. The method does not require collocation points for PDE solutions, instead, Ritz variational form is used, and hence, the energy functional is minimized in the loss function. For the functional, a numerical quadrature rule is used, and a stochastic gradient optimization algorithm (i.e. Adam optimizer) is applied for the minimization. The proposed approach is tested by solving the Poisson. They also tested transfer learning for better accuracies (Yu 2018). Similarly, Liao and Ming (2019) presented the deep Nitsche method that takes Nitsche's formulation as the loss function to tackle necessary boundary conditions instead of having a penalty as in the case of DRM.

In another work, a Deep Galerkin Method (DGM) has been presented to solve PDEs. In this method, instead of a linear combination of basis functions (as in the case of the Galerkin method), a deep neural network is used and the PDE residual is defined in the least-squares sense. This network is then trained to satisfy the initial and boundary conditions of PDEs and the differential operator at randomly sampled spatial points. The DGM method is tested on high-dimensional PDE which is found to solve PDE accurately with upto 200 dimensions (Sirignano and Spiliopoulos 2018). Similarly, in another approach, a Deep Petro-Galerkin Method (DPGM) is presented on a weak formulation which is based on variational principle. In DPGM, the deep neural network approximates the unknown functions, whereas, different approaches can be chosen for test functions. The method also uses the least-square method in its solution to PDEs (Shang et al. 2022).

Other works include the development of variational or a Petrov–Galerkin form of PINN i.e., variational PINN (VPINN) that contains a variational loss function and solve the weak (variational) problem. In this method, the trial space is the neural network space whereas the test space is the Legendre polynomials space. This method helps in reducing the training cost with increased accuracy than conventional PINN. In addition, the integral in the variational formulation is computed through the quadrature points (Kharazmi et al. 2019). Extending the VPINN, a method named hp-variational physics-informed neural networks (hp-VPINNs) has also been proposed that allows the hp-refinement via domain

decomposition, and the trial and test spaces are represented by the neural networks and piecewise polynomials respectively. With integration-by-parts, the proposed method is able to perform better in rough solutions such as sharp changes, steep solutions, and singularities. Also, it demonstrated advantages in both the training cost and accuracy (Kharazmi et al. 2021).

To this end, a summary of the major developments in PIML (with the major focus on the collocation-based approach) including the architectural and function changes is created that is shown in Fig. 5 and a pseudo code for a typical PINN is presented in Table 1. Moreover, a detailed description of all the major developments is presented in Table 2.

Table 1 Pseudocode for a typical PINN for continuous solution (modified from Kollmannsberger et al. 2021)

```

def Initialization:

    • Initial and boundary conditions

    • Collocation points with the appropriate sampling strategy

    • Type of network and network architecture (layers, neurons, activation functions, etc.)

    • Set optimizer hyperparameters

def Training:

    • Compute  $F_{pred}$ 

    • Compute derivatives i.e.  $F_{xx}$ ,  $F_{xxx}$ ,  $F_y$ ,  $F_{yy}$ ,  $F_t$  and so on

    • Compute defined equation residual

    • Compute function MSE errors i.e. initial conditions, boundary conditions, residual, and data error

    • Evaluate loss function

    • Update based on the selected optimizer

    • End

def optimization:

    for all Adam-epochs do

        run Training with Adam optimizer

    end

    for all L-BFGS-epochs do

        run Training with L-BFGS optimizer

    end

```

Table 2 Recent developments in physics-informed machine learning workflows in various disciplines in science and engineering (year-wise)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Raissi et al.	Loss function includes PDE residual loss as well	L-BFGS	Mean squared error (MSE)	Forward and Inverse	<i>Data-driven solution</i> (1) Schrödinger equation	(1) Tackle non-linear problems without prior assumptions, linearization, or local time-stepping	Raissi et al. (2019), Raissi and Karniadakis (2018)
	Simple FF architecture with hyperbolic tangent activation functions				(2) Allen–Cahn equation	(2) Limited low-dimensional applications	
	No additional regularization					(3) No generalized network setting	
Mao et al.					<i>Data-driven discovery of partial differential equations</i>		
	Loss function includes PDE residual loss as well	Adam and L-BFGS	MSE	Forward and Inverse	(1) Navier Stokes equation (2) Korteweg–de Vries equation–Burger’s equation	(1) Clustered training points provides more accurate solution than random or uniform points	Mao et al. (2020)
	Feed-forward neural network is used				1D and 2D Euler’s equations	(2) Difficulties in training sampling points without a priori knowledge of the solution	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Zhang et al.	Loss function includes PDE Loss and Data Loss Arbitrary polynomial chaos (aPC) is integrated with PINN (network is termed as NN-aPC)	Adam	MSE	Forward and inverse	Forward stochastic Poisson equation with driving force uncertainty Inverse stochastic elliptic and linear diffusion–reaction problem with diffusivity and forcing terms as random processes respectively	(3) Applications to 1D and 2D PDE are presented only (1) Solved stochastic partial differential equations (SPDE) with random parameters (2) Random parameters are introduced as stochastic processes (3) Total uncertainty including the parametric uncertainty from the various random parameters and approximation uncertainty from the DNN is introduced (4) Arbitrary polynomial chaos (aPC) is integrated with PINN to solve the stochastic problem since it helps in developing an orthogonal basis from random space without the assumption requirement of data distribution (5) Applied active learning iterative method to improve prediction accuracy	Zhang et al. (2019)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Zhu et al.	Compared CNN based approach with the FC-NN approach Loss includes PDE residual, IC/BC loss, and mixed residual.	Adam and L-BFGS	MSE	Forward	Darcy flow	(6) The dropout is used to handle the uncertainty in the aPC node of the PINN (7) Dimensionality reduction through principal component analysis (PCA) is applied to generate uncorrelated random variables in stochastic sampling space (8) Suffers "curse of dimensionality" while dealing high dimensional data (9) Limitations in solving strong non-linear time-dependent stochastic equations	Zhu et al. (2019)
						(1) Solved both deterministic and stochastic PDEs while considering CNN encoder-decoder architecture without simulation data (2) CNN based approach was found to be computationally efficient as compared with FC-NN in solving deterministic PDEs (3) Approach provides better extrapolation in predictive mode (4) A flow-based conditional generative model is also proposed with comparable predictability under uncertainty (5) Generalization performance is found to be better than the data-driven methods	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Meng et al.	NN is the combination of four networks including one low-fidelity DNN, two high-fidelity DNNs, and a PDE-induced DNN Loss function includes low-fidelity network loss, high-fidelity network loss, and PDE loss	Xavier's initialization and L-BFGS method	MSE	Inverse	Hydraulic conductivity for nonlinear unsaturated flow	(6) High dimensional problems can be solved through the CNN approach with accuracy with the requirement of the order of magnitude few parameters (1) It is a multi-fidelity PINN with four DNNs	Meng and Karniadakis (2020)
					Reaction models for reactive transport	(2) Hyperbolic tangent function is used as an activation function	
						(3) Applicable for high-dimensional as well as multi-fidelity problems (4) Applicable to both nonlinear and linear correlation (5) Can approximate both discontinuous and continuous functions	
Fraces et al.	Loss function includes PDE residual loss as well	Adam and L-BFGS	MSE	Forward and Inverse	Buckley–Leverett equation	(1) Used random sampling, fixed location sampling, and early time sampling	Fraces et al. (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
	Simple a feed-forward multilayer perceptron (FF-MLP) with a non-linear activation function					(2) Generative Adversarial Networks (GAN) application added	
	Generative Adversarial Networks (GAN) are also tested					(3) Introduced diffusion and gravity terms in the network	
						(4) 2D water–oil displacement case is also presented	
						(5) Limited to 2D transport equation only	
						(6) Approach is problem-specific	
Yang et al.	Includes multiple DNNs, PDE solution module, and Adversarial training module	Adam	–	Forward, inverse, and mixed problems	High-dimensional elliptical SPDE	(1) Used GAN specifically Wasserstein GANs with gradient penalty (WGAN-GP) to solve SPDE	Yang et al. (2020)
	Includes generator and discriminator loss					(2) This method does not suffer from the curse of dimensionality, hence, can be appropriate for high-dimensional data	
						(3) Impact of noise is also tested	
						(4) Also compared WGAN-GP with vanilla GAN and found that vanilla GAN is not suitable for SPDE having deterministic boundary conditions	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Zhang et al.	Dynamically orthogonal (DO) and bi-orthogonal (BO) methodologies are integrated with PINN (NN-aPC)	Adam	MSE	Forward and inverse	Linear stochastic advection equation	(5) Issues of overfitting were also observed during training (6) The computational cost is relatively higher than a simple FFNN (1) Solved more complex non-linear time-dependent SPDE	Zhang et al. (2020)
	Loss function includes PDE loss, IC/BC loss, data loss, DO/BO loss, and additional regularization terms loss				Non-linear stochastic burger problem	(2) Employed time-dependent Karhunen-loève (KL) decomposition to evolve time-dependent basis of stochastic coefficients and modes	
					Non-linear diffusion reaction equations	(3) Tested two different methodologies i.e. bi-orthogonal (BO) methodology for spatial and stochastic basis and dynamically orthogonal (DO) for the spatial basis to represent static and dynamic constraints in the representation respectively (4) The method has limited accuracy because of the complex non-convex optimization problem	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Dwivedi et al.	Combined extreme learning machine (ELM) with PINN Loss function includes PDE loss and IC/BC loss Shallow network is used	–	MSE	Forward	Steady and unsteady advection–diffusion problem	(5) Require high computational cost because of integration of BO and DO methods (1) The proposed method i.e. physics informed extreme learning machine (PIELM) is a combination of ELM and PINN (2) PIELM uses linear differential operators and is valid for linear PDEs only (3) The proposed method is data efficient and fast with high accuracy as compared with regular PINN (4) PIELM is inaccurate in predicting sharp and discontinuous solutions (5) Also proposed distributed PIELM (DPIELM) to enhance the efficiency of PIELM and fixes issues of sharp gradients of PIELM (6) DPIELM distributes a single computational domain into multiple sub-domains where simple neural networks represent PDE in each sub-domain (7) Limitations in handling complex geometries	Dwivedi and Srinivasan (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Almajid et al.	Loss function includes PDE residual loss as well Fully connected neural network was used	Adam and L-BFGS	MSE	Forward	Buckley-Leverett equation	(1) Low and high mobility ratios were used (2) Study is based on diffusion parameters, observed data, and multiphase trainable parameters (3) Limited to 1D Buckley-Leverett problem	Almajid and Abu-Alsaud (2020)
Sun et al.	Loss function includes PDE residual only (with Newman BC loss) Initial and boundary conditions are encoded in the output layer by construction (IC and Dirichlet BC) -Data-free Deep neural networks (No simulation data)	Adam	MSE	Forward	Navier Stokes equation for: (1) Circular pipe flow (2) Blood flow	(1) Instead of penalizing IC/BC, they enforced them by construction (2) Mass and momentum conservation is included in the loss function (physics-based loss) (3) No numerical discretization is utilized (4) Also presented a study on different activation functions in the proposed method (5) Applications to simple flow are presented	Sun et al. (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Kadeethum et al.	Loss function includes additional PDE loss Fully connected multilayered neural network was used	Combination of ADAM, SGD, and L-BFGS	MSE	Forward and Inverse	Diffusivity equation and Biot's equation	(6) Data-driven approach is found to be slightly more accurate than the proposed method, however, it requires large data which in the case of large 3D simulation is computationally expensive (7) Only low-dimensional applications are presented (1) Additionally, worked on hyper-parameter choices, training sizes, stochastic variations (2) Noise impact is also investigated	Kadeethum et al. (2020)
Guo et al.	Loss function includes PDE residual loss as well Feedforward Neural Network	ADAM, SGD, L-BFGS, RMSprop	MSE	Forward	Wave equation	(3) Limited application to low-dimensional problem (4) Generalization is missing (1) Training data includes initial and boundary conditions and random Spatio-temporal data (2) Regularization takes the L2 norm form (3) Low-dimensional (1D PDE) application only	Guo et al. (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Voytan et al.	Loss function terms are constituted by initial and boundary conditions and differential operator Fully connected neural network	L-BFGS	MSE	Forward	Scalar wave equation: (1) Homogeneous	(1) Extended wave equation to 2D (2) For inhomogeneous conditions, spatially varying velocity and absorbing boundary conditions are utilized (3) Used adaptive L4-L2 norm (4) Slow training is observed (5) Applications on complicated velocities are missing	Voytan and Sen (2020)
Kumar et al.	Loss function includes PDE residual loss Dense deep neural network	L-BFGS	MSE	Data-driven PDE discovery approach	(2) Inhomogeneous Profiling of subsurface velocity	(1) Training dataset includes the data obtained through Python Seismic Imaging Toolbox (PySIT) and real subsurface receiver data from the Volve field (2) Limited training data 7.5% of the actual. (3) Tried PINN on real applications (4) Problem-specific approach	Kumar et al. (2020)
Jagtap et al.	Loss function includes PDE residual loss as well Dense deep neural network	Adam	MSE	Forward and inverse	Burgers equation	(1) Employed adaptive activation function instead of fixed activation function (2) This adaptive function allows fast convergence rates and increased accuracy	Jagtap et al. (2020a)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Jagtap et al.	Global adaptive activation function is employed which includes an adaptive hyperparameter with a scaling factor				Klein-Gordon equation	(3) Solved low-dimensional PDE	Jagtap et al. (2020b)
	Loss function includes PDE residual loss	Adam	MSE	Forward and Inverse	Non-linear discontinuous problem	(1) Improved training speed as compared to global adaptive functions	
	Dense deep neural network				2D inviscid Burgers' equation	(2) Added computational cost	
	Local adaptive activation function (LAAF) is employed including neuron-wise and layer-wise functions				2D Poisson equation	(3) Solved low-dimensional PDE	
	Slope-based recovery term is added to the loss function				Standard DL benchmark problems		

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Jagtap et al.	Loss function includes neural network loss, PDE residual loss, and interface conditions (flux continuity conditions) Feed-forward neural networks Domain Decomposition based PINN framework (space only)	Adam	MSE	Forward and Inverse	kdV equation Burgers equation Eulers' equations Navier–Stokes equation	(1) Proposed PINN for conservation laws i.e. conservative PINN (cPINN) with spatial domain decomposition (2) Local adaptive activation functions are used for fast convergence (3) Slope recovery terms are also used (4) Domain decomposition provides easy parallelization (5) 1D and 2D problems are solved using this approach (6) Offers better computational cost as compared to the traditional PINN version (7) Domain decomposition is not possible in any arbitrary way (in time and space)	Jagtap et al. (2020c)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Jagtap et al.	Loss function includes neural network Loss, PDE residual Loss, and interface conditions (residual continuity conditions) Feed-forward neural networks Generalized Domain Decomposition (in space–time) Deploy multiple neural networks in smaller domains	Adam	MSE	Forward and Inverse	Linear advection equation Viscous Burgers equation 2D and 3D Poisson's equation Compressible Euler's equation 2D elasticity equations	(1) Domain decomposition is possible in any arbitrary way (in time and space) (2) Parallelization is offered in both time and space (3) This updated method from cPINN is named extended PINN (XPINN) (4) Applied on 1D to 3D problems (5) Can be extended on any PDE (6) Improved computational cost with parallel implementation (7) Usefulness in dynamic interface problems	Jagtap and Karniadakis (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Tartakovsky et al.	Loss function includes data loss and PDE residuals including boundary conditions Multiple DNNs	L-BFGS-B	MSE	Ill-posed	(1) Parameter estimation: Linear diffusion equation (2) Unknown physics: Non-linear diffusion equation (3) Specific application: Hydraulic conductivity estimation	(4) Limitations in high-dimensional complex flows (not tested in the presented work) (1) Parameter estimation is found to be more accurate than the MAP estimation method (2) Accurately estimates constitutive relationship (state-dependent diffusion coefficient) (3) PINN improves DNN parameter estimation accuracy by 50%	Tartakovsky et al. (2020)
He et al.	Loss function includes data loss, PDE residual loss, and boundary conditions loss	Adam and L-BFGS-B	MSE	Data assimilation	Assimilation of space-dependent hydraulic head, conductivity, and concentration fields	(4) Confirmed that the PINN is less beneficial for the problems with large data (5) No noise assumptions (6) Regularization effect is not studied (1) Multiphysics measurements in transport problem	He et al. (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Gao et al.	DNNs represent the space-dependent head, conductivity, and concentration fields					(2) Included sparse measurements	
	Advection–dispersion and Darcy equations are used alongside the observed data					(3) Tested DNN, data size, and parameter field complexity	
						(4) Accuracy is better than data-driven DNN, especially with the sparse data	
						(5) Parameter estimation accuracy increases with the addition of more variables inverted jointly	
Gao et al.						(6) Computational cost is higher for training	
	Loss function includes equation and boundary condition loss	Adam	MSE	Forward	Heat equation	(1) Proposed physics-informed geometry-adaptive convolution neural network (PhyGeoNet) to solve parametric PDEs with irregular geometry and non-uniform grids	Gao et al. (2021)
	CNN is used as a solver				Navier Stokes equation Poisson equation	(2) Irregular coordinates are transformed into a rectangular domain	
						(3) Loss function is constructed by reference domain PDEs through finite difference convolution kernels	
						(4) Padding operations are used to strictly enforce boundary conditions	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Geneva et al.	Used convolution encoder-decoder network	Adam	MSE	Forward	1D Kuramoto–Sivashinsky (K–S) equation	(5) Different parabolic and elliptic equations are tested	Geneva and Zabaras (2020)
	Loss function includes the difference minimization between discrete numerical time-integrator of PDE and the model's prediction				1D viscous and 2D coupled Burgers' equation	(6) Application to parametric and non-parametric solutions are explored	
	Periodic boundaries are enforced in the loss function					(7) The proposed approach performed better in terms of accuracy and computational time compared to FC-NN conventional PINN	
						(8) This approach cannot deal with dynamic systems	
						(9) Limitations in handling more complex structures/domains	
					(1) Deep auto-regressive dense encoder-decoder network is used to solve transient PDEs		
					(2) The model is extended to the Bayesian framework through Gaussian stochastic weight averaging for aleatoric (observation noise) and epistemic (model uncertainty) uncertainty quantification		
						(3) The proposed modules are named the auto-regressive dense encoder-decoder model (AR-DenseED) and Bayesian AR-DenseED (BAR-DenseED)	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Yang et al.	Used Bayesian neural network Posterior includes Variational inference (VI) or Hamilton Monte Carlo (HMC) methods	Adam	-	Forward and inverse	1D linear and non-linear Poisson equation	(4) Methodology is implemented over 2D coupled nonlinear PDEs and chaotic systems (5) Drawback of using CNN is its requirement for the discretization of both temporal and spatial derivatives	Yang et al. (2021)
					1D flow equation	(1) Solve PDEs with noisy data with uncertainty quantification	
						(2) The source and boundary terms are obtained through the noisy data	
					2D non-linear Allen-Cahn equation	(3) It includes a Bayesian framework containing a Bayesian neural network with a prior containing PDE and posterior estimate through Variational inference (VI) or Hamiltonian Monte Carlo (HMC) and named networks as B-PINN-HMC and B-PINN-VI	
					1D/2D non-linear diffusion-reaction system	(4) Compared B-PINN-HMC with PINN and determined that the PINN is less accurate with overfitting issues than B-PINN-HMC	
					6D source inversion problem	(5) For B-PINN posterior estimation, the HMC is found to be more suitable than VI	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Kumar et al.	Combined graph neural network (GNN) with Neural ODE	-	MSE	Forward	1D and 2D Burger's equation	(6) They also replaced BNN as a prior with KL expansion along with deep normalizing flow (DNF) or HMC as posterior and found that the KL is more accurate but cannot be extended to high dimensions (1) Proposed graph attention differential equation (GrADE) that combines graph neural networks (GNN) specifically graph attention (GAT) with Neural ODE (Chen et al. 2018) for solving time-dependent PDEs (2) Use of GNN helped in handling irregular domain (3) The neural ODE models the temporal domain and GNN models the spatial domain (4) Boundary conditions are defined on graph nodes (5) Obtained better generalization in future time steps (6) Also used depth refinement training strategy for better accuracy and reducing training time	Kumar and Chakraborty (2021)
Iakovlev et al.	Used graph neural network (GNN) to solve PDEs	Rprop	-	Forward	Convection-diffusion equation	(1) GNN specifically message passing neural networks (MPNNs) has been used to learn PDE from sparse data	Iakovlev et al. (2020)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error func-tion	Problem nature	PDEs solved	Features and limitations	References
Zhang et al.	Incorporated the minimum energy criteria into the loss function	Adam	MSE and averaged R2	Free of ground truth data	Heat equation	(2) The developed methodology includes continuous-time unknown PDE learning, free-form spatial domain, and initial/boundary conditions	Zhang and Gu (2021)
					Burger's equation	(3) The model can perform better for PDE systems with high sparse data	
					Digital Materials—1D Problem including displacement responses (Dirichlet boundary condition) and bending responses (Neumann boundary condition)—2D digital material deformation prediction	(4) The methods of lines (MOL) is used to solve the equations	
						(1) Used minimum energy criteria instead of governing equations	
						(2) A hinge loss was added on the Jacobian to constrain the model restricting the erroneous deformation gradient resulting from logarithmic strain (non-linear in nature)	
						(3) Limited 1D and 2D applications	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Rodriguez-Torrado et al.	Loss function includes PDE residual only Initial and boundary conditions are enforced in the network architecture rather than the loss function (compared to previous PINN methods) Combined attention mechanisms with general recurrent units (GRU)	Adam	MSE	Forward	Buckley-Leverett problem	(1) Focus is concentrated on network architecture rather than residual regularization (2) Combination of attention mechanisms and general recurrent units (GRU) (3) High-quality solutions beyond and inside the training set (4) the First method utilized attention mechanisms for solving non-linear PDEs for hyperbolic problems with discontinuity (5) Limited complexity	Rodriguez-Torrado et al. (2021)
Ren et al.	Loss function includes aggregated discretized PDE residual.	Adam and L-BFGS	RMSE	Forward and Inverse	λ - ω and FitzHugh-Nagumo reaction–diffusion equations, and 2D Burgers' equations	(1) Accurate in extrapolation and generalization	Ren et al. (2021)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
	IC/BCs are hard-coded into the network Encoder-decoder convolutional long short-term memory (ConvLSTM) network Need no labeled data					(2) No labeled data is required (3) Contains high-order finite-difference-based spatiotemporal filtering for residual loss (4) ConvLSTM learns temporal evolution and extracts low-dimensional spatial features (5) It is a mesh-based approach (spatiotemporal mesh) and is limited to regular grids	
Waheed et al.	Loss function includes the eikonal equation Deep neural network	Adam	MSE	Forward	Solution of the eikonal equation	(1) No labeled dataset (2) Locally adaptive inverse tangent activation function is used (3) Adaptive weighting is assigned in the loss function	Waheed et al. (2021)
Shukla et al.	PDE residual is added in the loss function Feed-forward neural network	Adam and L-BFGS	MSE	Ill-posed inverse problem	Quantification of Nickel microstructure properties using ultrasound data	(4) Limited to 2D model (1) Used second-order PDE defining out-plane and in-plane displacement (2) Slow convergence at some frequencies	Shukla et al. (2021a)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Eivazi et al.	Employed global adaptive activation functions						
	Used domain boundary data with RANS equations for the learning process	Adam and BFGS	MSE	-	Reynolds-averaged Navier–Stokes (RANS) equation for incompressible turbulent flows with	(1) For laminar flow, the prediction error is less than 1%	Eivazi et al. (2021)
	Loss function includes boundary data loss and PDE residual.				(1) adverse-pressure-gradient (APG) boundary layer (2) zero-pressure-gradient (ZPG) boundary layer (3) periodic hill (4) turbulent flows over a NACA4412 airfoil	(2) In turbulent flow, the accuracy is better even for the Reynolds-stress components (3) L2-norm of errors is used for the accuracy	
Wright et al.	Novel Physics aware training approach which executes a back-propagation algorithm for physics input–output transformation	SGD	MSE	Physical systems	Tested PNN approach on the mechanical, electrical, and optical physical system	(1) Backpropagation algorithm is used in training physical systems to execute deep neural networks	Wright et al. (2021)

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Li et al.	Physical systems are fed into the forward pass					(2) Physical systems are embedded in the backpropagation process	
	Physical system gradient is calculated using a differentiable digital model w.r.t controllable parameters					(3) Forward pass is executed through the physical system	
	Added PDE constraint to operator learning	Adam and L-BFGS	MSE	Forward and inverse	2D Darcy flow	(1) Used one neural operators model that can solve both equations solving problems and operator learning problems	Li et al. (2021a)
	Loss function includes both the data and PDE loss				1D Burgers' equation	(2) Proposed two PINO workflows: forward PINO and backward PINO	
	Two phases: (1) Solution operator pre-training and (2) Test-time optimization				Navier-Stokes Equation: Chaotic Kolmogorov flow, Long temporal transient flow, Lid cavity flow	(3) Proposed exact gradient method for derivative calculation	
						(4) Overcome PINN challenges of (i) information propagation from the initial and boundary condition to future times or unseen interior parts (ii) PDE convergence on multiscale or high-frequency PDEs structure	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Krishnapriyan et al.	Loss function includes data loss, residual loss, and boundary loss Sequence-to-sequence and curriculum regularization strategies are proposed Feed-forward neural network is used	L-BFGS	MSE	Forward	Convection problem Reaction–diffusion problem	(5) Better generalization with no or fewer training data than FNO (6) Utilize a function-wise optimization approach instead of pointwise (in the case of PINN) (7) Much faster and more accurate compared to PINN (8) Has accuracy and complexity trade-off	Krishnapriyan et al. (2021)
						(1) Analyzed vanilla/regular PINN and found that it fails on challenging regime (chaotic) problems	
						(2) Analyzed loss landscape of PINN and found that the addition of PDE soft constraint regularization results in hard optimization	
						(3) Also investigated that loss landscape problem can be alleviated by reducing regularization parameter, however, the predictions are poor (4) Two methods are proposed to deal with the above challenges i.e. curriculum regularization and sequence-to-sequence learning	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
Wang et al.	A weighted residual loss is introduced with the causality parameter	Adam	MSE	Forward	Kuramoto–Sivashinsky equation	(5) In curriculum regularization, small coefficients for PDE constraints are imposed initially that are increased progressively during training. Such a setting allows NN to train easier constraints before tackling target constraints	Wang et al. (2022)
	Feed-forward neural network is used				Lorenz system	(6) In sequence-to-sequence learning, instead of training on full space–time, the NN is trained on specific smaller time intervals (7) Both the approaches improved the training performance by a magnitude of 1–2 orders as compared to the regular PINN (1) Deals with initial/boundary condition causality issues while solving continuous PDEs through PINN (2) Re-formulated the loss function of PINN to respect causality in a chaotic system	

Table 2 (continued)

Author	Approach	Optimization and optimizers	Error function	Problem nature	PDEs solved	Features and limitations	References
					Navier Stokes equation	(3) The weighted residual loss is introduced to allow the previous residual loss to minimize first before later time residuals (4) Training is sensitive to the causality parameter i.e. smaller value reduces minimization capabilities of latter temporal residuals, while, a very high value presents a challenge of solving a difficult optimization problem (5) Annealing strategy is hence applied to adjust causality parameter (6) The proposed framework is able to solve chaotic Kuramoto–Sivashinsky equation, chaotic Lorenz system, and Navier stokes equation in turbulent regime	

4.1 PIML applications in various disciplines in the science and engineering

With the development of the PIML workflow, various scholarly peer-reviewed research papers have shown the capability of this workflow in solving various partial differential equations utilized in various science and engineering field especially relating to fluid flow, seismic analysis, thermal profiling, and energy conservation. A description of some equations solved/utilized by the machine learning algorithm is presented in Table 3. In this section, we summarize some of the key applications and takeaways from implementing PIML to solve various science and engineering data-driven problems.

4.1.1 Non-linear partial differential equations

Raissi et al. (2018) presented physics-informed neural networks (PINN) for the training of neural networks governing physical laws described by non-linear partial differential equations. Their framework includes solutions on continuous and discrete-time models with problems including two distinct classes: data-driven solution and data-driven discovery of partial differential equations. The general PDE in their work is presented as (Raissi et al. 2019):

$$u_t + \mathcal{N}[u; \lambda] = 0, x \in \Omega, t \in [0, T], \quad (2)$$

where $u(t, x)$ represents the hidden solutions with Ω being the $\mathbb{R}^D$ the subset, and $\mathcal{N}[\bullet; \lambda]$ denotes a non-linear operator that is parameterized by λ . Their methodology includes the same as presented in Eq. 1 i.e. reducing the loss (MSE) for shared parameters between the neural network and PDE. In addition, their approach works on the same automatic differentiation (Baydin et al. 2018) by taking the derivative with respect to space–time input coordinates in a PDE.

In their approach, the data-driven solution includes the solution of $u(t, x)$ while the fixed-known parameter λ . In this domain, they presented their methodology on two equations that are Schrodinger equation (A equation widely used for understanding quantum mechanical systems including Bose–Einstein condensates and wave propagation) in the continuous case while the Allen–Cahn equation (applied usually to the reaction–diffusion system including the multicomponent phase separation in alloy system) in the discrete case. They utilized classical Runge–Kutta time-stepping in the discrete case. On the contrary, the data-driven discovery of partial differential equations includes the estimations of λ best describing the observed data. In this solution, they used classical incompressible fluid flow equation i.e. Navier Stokes equation which is highly regarded for fluid flow applications in science and engineering. For a discrete solution, in this case, they applied their methodology to the Korteweg–de Vries equation which deals with the evolution of waves (1D) in different physical systems. Their obtained results are presented in Fig. 6a and b. All the results are compared with the exact solutions of the PDEs. The results indicated that the neural network with physical constraints embedded in the loss function made the training process effective with accurate predictions and PDEs parameter identification. This indeed shows the PINN's ability to obey the laws of physics during the predictions.

4.1.2 1D and 2D digital materials

Zhang et al. (2021), in their work, design PINN for digital material (DM) problems. Their workflow is consistent with the general workflow presented earlier. They incorporated the minimum energy criteria into the loss function of the neural network. Also, they added a hinge loss on the Jacobian to constrain the model restricting the erroneous deformation gradient resulting from logarithmic strain (non-linear in nature). For the 1D problem, they tested their methodology with composite beam displacement responses (Dirichlet boundary condition) and bending responses (Neumann boundary condition) under different material configurations (Zhang and Gu 2021). Their generated PINN was found to be consistent with the numerically simulated results along with comparable results with the supervised ML model under different training data (see Fig. 7a).

For 2D DM, they used PINN and trained the model for 50 epochs with Adam optimizer on 4500 randomly generated sets. Their trained PINN showed a 0.021 testing error with an R^2 of 90.48. Figure 7b shows the comparison between the simulated and predicted shape of 2D material by PINN. The result shows a good agreement between both these shapes showing the effective implementation of PINN of 2D DM.

4.1.3 Applications for transport in porous media: Buckley-Leverett problem

The Buckley-Leverett problem is one of the classical problems in the energy industry that governs immiscible displacement in the porous media. Specifically, it calculates the rate at which the water moves in the porous media. The equation, in its general form, can be presented as (Buckley and Leverett 1942):

$$\frac{\partial S_w}{\partial t} = \frac{\partial}{\partial x} \left(\frac{Q}{\phi A} f_w(S_w) \right) = \phi \frac{\partial S_w}{\partial x} \quad (3)$$

where S_w is the water saturation, Q is the flow rate, A is the cross-sectional area, ϕ is the porosity, and f_w is the water fractional flow.

Fraces et al. (2020) presented the application of PINN to the Buckley-Leverett problem. Their approach solves PDE of the form:

$$\mathcal{R}(u_t, t, x, u, u_x, u_{xx}, \dots) = 0 \quad (4)$$

In their approach, the unknown solution u can be represented by a feed-forward multi-layer perception (FF-MLP) with non-linear activation function g , weight w , and bias b as presented in Eq. 5. In their approach also, automatic differentiation is used to solve PDE.

$$u \approx g(W_o \times (g(W_1 \dots) + b_o)) \quad (5)$$

The Buckley–Leverett problem is considered an inverse problem, in this approach, with the determination of PDE hidden parameters consistent with the observed data. The PDE residual $\mathcal{R}$ for Buckley-Leverett is defined as:

$$\mathcal{R} = \frac{\partial \hat{S}}{\partial t} + \frac{\partial f}{\partial \hat{S}} \frac{\partial \hat{S}}{\partial x} \quad (6)$$

The loss is calculated on the general workflow determined earlier in Eq. 1 by combining observed data loss with the residual loss of PDE. A total of 1000 training data points are

Table 3 Integration of different physical laws with machine learning algorithms for various science and engineering applications

PDE name	Equation with description	Discipline (major applications)	References
<i>Schrodinger equation</i>	A non-linear Schrödinger equation with periodic boundary conditions is given as (Raissi et al. 2019): $ih_t + \frac{1}{2}h_{xx} + h ^2h = 0$ Where, $h(t, x)$ is the complex-valued solution	Quantum mechanics Optics	Raissi et al. (2019)
<i>Allen-Cahn equation</i>	A nonlinear form with periodic boundary conditions is given as (Raissi et al. 2019): $u_t - 0.0001u_{xx} + 5u^3 - 5u = 0$ Where, $u(t, x)$ is the complex-valued solution	Reaction-diffusion system Material science Phase separation Biological system	Raissi et al. (2019) and Matthey and Ghosh (2022)
<i>Navier Stokes equation</i>	Navier Stokes (N-S) in 2-dimension can be represented as (Raissi et al. 2019): $u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy})$ $v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy})$ where u and v are x- and y-components of the velocity field respectively, and $p(t, x, y)$ is the pressure. λ is the parameter A more general incompressible, Steady, Newtonian, N-S equation is (Sun et al. 2020; Arzani et al. 2021): $\rho u \cdot \nabla u = -\nabla p + \mu \nabla^2 u$ $\nabla \cdot u = 0$ where u is the velocity vector, ρ is the density, and μ is the dynamic viscosity	Fluid flows Medical science/Biomedical engineering Aerospace engineering Maritime engineering Geosciences	Raissi et al. (2019), Sun et al. (2020), Arzani et al. (2021), Oldenburg et al. (2022), and Cai et al. (2022)
<i>Korteweg-de Vries equation</i>	The Korteweg-de Vries (kdV) equation in 1-dimension can be defined as (Raissi et al. 2019; Guo et al. 2020): $u_t + \lambda_1 uu_x + \lambda_2 u_{xxx} = 0$ Where $u(t, x)$ is the complex-valued solution, and λ is the parameter	Water wave Acoustic wave Ion-phonon waves Fluctuations in biological and physical systems	Raissi et al. (2019) and Guo et al. (2020)

Table 3 (continued)

PDE name	Equation with description	Discipline (major applications)	References
<i>Burgers' equation</i>	A 1-dimension Burgers' equation can be defined as (Raissi et al. 2019; Guo et al. 2020; Jagtap et al. 2020a): $u_t + \lambda_1 u u_x - \lambda_2 u_{xx} = 0$ In 2-dimensions, it can be represented as (Ren et al. 2021): $u_t + u \bullet \nabla u - \nu \Delta u = 0$ Where u is the velocity, ν is the viscosity coefficient, Δ is the Laplacian operator, and λ is the parameter	Applied mathematics Gas dynamics Acoustic Traffic flow Fluid mechanics and dynamics	Raissi et al. (2019), Guo et al. (2020), Jagtap et al. (2020a) and Ren et al. (2021)
<i>Euler's equation</i>	Euler's equation can be represented as (Mao et al. (2020): $\partial_t U + \nabla \cdot f(U) = 0$ In 1-dimension: $U = \begin{pmatrix} \rho \\ \rho u \\ \rho E \end{pmatrix}, f(U) = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ u(\rho E + p) \end{pmatrix}$ In 2-dimensions: $U = \begin{pmatrix} \rho \\ \rho u_1 \\ \rho u_2 \\ \rho E \end{pmatrix}, f = (G_1, G_2), \text{ with } G_i(U) = \begin{pmatrix} \rho u_i \\ \delta_{i1} p + \rho u_1 u_i \\ \delta_{i2} p + \rho u_2 u_i \\ \rho u_i + \rho u_i E \end{pmatrix}, i = 1, 2$ Where p is the pressure, ρ is the density, u is the velocity with u_1, u_2 are its components in the x and y directions, δ_{ij} is the Kronecker delta, and E is the total energy	Aerodynamic flows Shock wave	Mao et al. (2020)
<i>Buckley-Leverett equation</i>	The Buckley-Leverett equation can be defined as: $\frac{\partial S_w}{\partial t} = \frac{\partial}{\partial x} \left(\frac{Q}{\phi A} f_w(S_w) \right) = \phi \frac{\partial S_w}{\partial x}$ Where S_w is the water saturation, Q is the flow rate, A is the cross-sectional area, ϕ is the porosity, and f_w is the water fractional flow	Fluid transport in porous media Geoscience	Fraces et al. (2020), Almajid and Abu-Elseud (2020) and Rodriguez-Torradó et al. (2021)

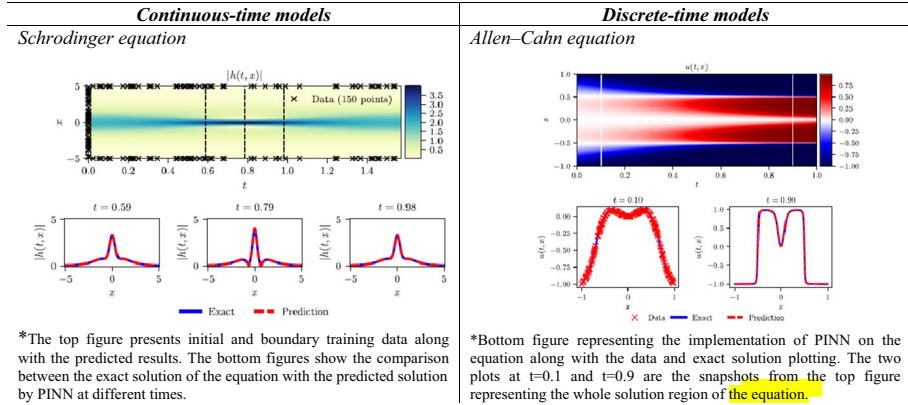
Table 3 (continued)

PDE name	Equation with description	Discipline (major applications)	References
<i>Diffusivity equation</i>	The non-linear diffusivity equation is represented as (Kadeethum et al. (2020): $\phi c_p P_i - \nabla \cdot \mathcal{N}[\kappa](\nabla p - \rho g) = g$ Where c_i is the total compressibility, p is the pressure, ϕ is the porosity, g is the gravity, ρ is the density, κ is the hydraulic conductivity, and $\mathcal{N}[\bullet]$ is the non-linear operator	Fluid flow in porous media Geoscience	Kadeethum et al. (2020)
<i>Wave equation</i>	The 1-dimensional wave equation can be written as (Guo et al. 2020 ; Voytan and Sen 2020): $u_{tt} = cu_{xx}$ Where c is the velocity of wave propagation, and u is the function of x and t	Radio communication Seismicity Acoustic wave propagation	Guo et al. (2020)
<i>Klein–Gordon equation</i>	An in-homogeneous Klein–Gordon equation is defined as (Jagtap et al. 2020a): $u_{tt} + \alpha \Delta u + N(u) = h(x, t)$ Where u and h are functions, Δ is a Laplacian operator, and $N(u)$ is a non-linear term	Condensed matter physics Solid state physics Optics Quantum theory	Jagtap et al. (2020a)
<i>Helmholtz equation</i>	In 2-dimensions, the Helmholtz equation is defined as (Jagtap et al. 2020a): $\Delta u + k^2(u) = q(x, y)$ Where u and h are functions, Δ is a Laplacian operator, and $N(u)$ is a non-linear term	Acoustics Vibrations Electromagnetism	Jagtap et al. (2020a)
<i>λ-ω RD equation</i>	A 2-dimensional representation of the λ-ω RD equation is (Ren et al. 2021): $u_t = 0.1 \Delta u + \lambda(r)u - \omega(r)v$ $v_t = 0.1 \Delta v + \omega(r)u + \lambda(r)v$ $r = u^2 + v^2$ Where ω and λ are real functions given as $\lambda = 1 - r^2$ and $\omega = -r^2$, and v, u are field variables	Multiscale phenomena in biological and chemical processes such as self-organized patterns and turbulent behavior	Ren et al. (2021)

Table 3 (continued)

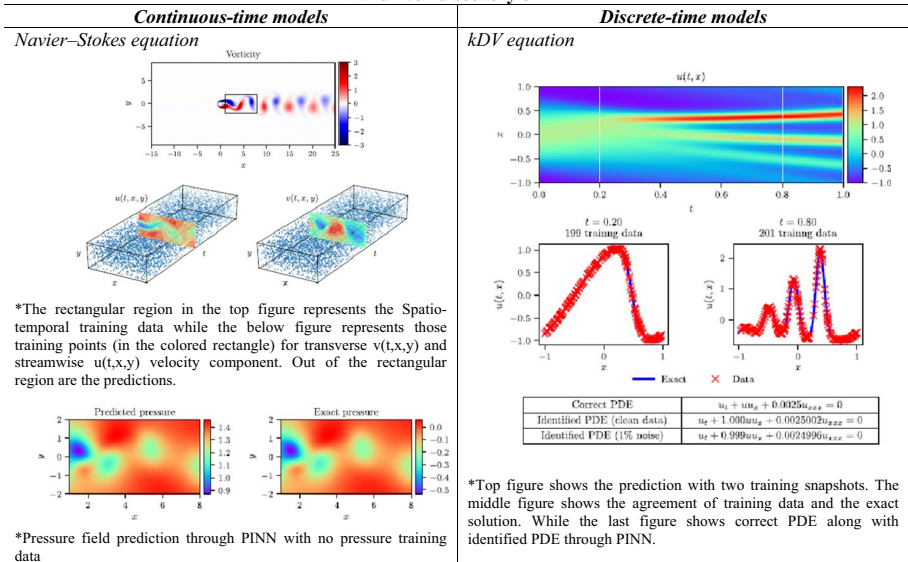
PDE name	Equation with description	Discipline (major applications)	References
<i>Eikonal equation</i>	The Eikonal equation is defined as (Waheed et al. 2021): $ \nabla T(x) ^2 = \frac{1}{v^2(x)}$ $T(x_s) = 0$, where $v(x)$ is the velocity, and $T(x)$ is the Euclidean distance or travel time between point-source x_s and any point x	Image processing Optics Robotics Computer graphics Seismology Manufacturing	Waheed et al. (2021)
<i>Darcy equation</i>	The Darcy equation can be defined as: $-\nabla \cdot (K(x)\nabla u(x)) = f(x)$ where $u(x)$ is the field variables of interest (hydraulic head/pressure), $f(x)$ is the source term, and $K(x)$ is the hydraulic conductivity	Hydraulic flow Subsurface flow Porous media	Zhu et al. (2019) and He and Tartakovsky (2021)
<i>Heat Equation</i>	The heat equation can be defined as: $u_t = D\nabla^2 u$ where u is the temperature field, D is the thermal diffusivity	Manufacturing Industrial engineering Process engineering Heat transfer include thermal convection, chemical vapor deposition crystal growth and solidification, welding, semiconductor design	Iakovlev et al. (2020), Cai et al. (2021b) and Zobeiry and Humfeld (2021)

The data-driven solution to PDE



a PINN approximations on data-driven solutions - Courtesy: Raissi et al. (2018) [40]

Data-driven discovery of PDE

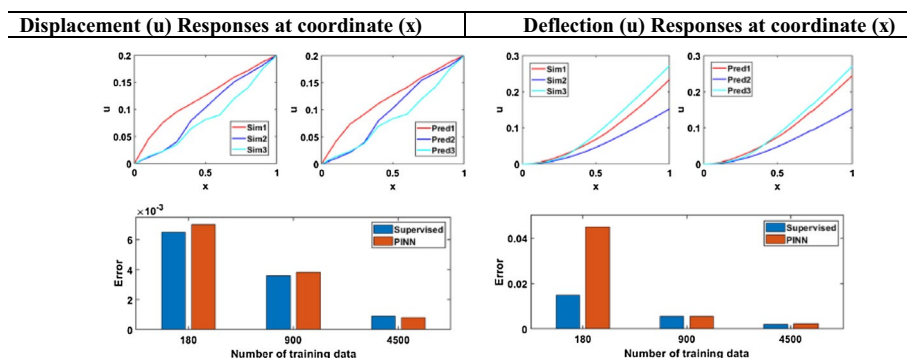


b PINN approximations on the data-driven discovery of PDE - Courtesy: Raissi et al. (2018) [40]

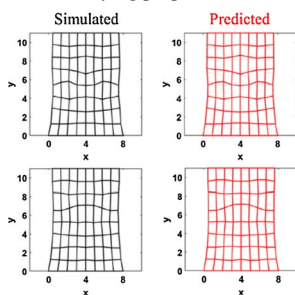
Fig. 6 **a** PINN approximations on data-driven solutions—Courtesy: Raissi et al. (2019). **b** PINN approximations on the data-driven discovery of PDE—Courtesy: Raissi et al. (2019)

used for PINN training. After training, it has been found that the physics-informed machine learning approach results in consistent and accurate results with physically consistent behavior with the training set data and the exact solution (refer to Fig. 8).

These results are based on random sampling. In addition, the authors further extended their work to fixed location sampling to replicate the behavior of multiple wells around an injector. They also tested their approach at early time sampling. In these cases too, they found the network prediction consistent with the observations and exact solutions with low mean squared errors (MSE).



a PINN for 1D digital material with varying properties – Courtesy: Zhang et al. (2021) [53]



b PINN application on 2D digital material deformation prediction - Courtesy: Zhang et al. (2021) [53]

Fig. 7 (a) PINN for 1D digital material with varying properties—Courtesy: Zhang and Gu (2021). **b** PINN application on 2D digital material deformation prediction—Courtesy: Zhang and Gu (2021)

Further extending their work, the authors presented Generative Adversarial Networks (GAN) (Goodfellow et al. 2014) application to the Buckley-Leverett problem inspired by the works of Yang et al. (2020) and Yang and Perdikaris (2019). This approach again results in a consistent match with the observed data and the exact solution. In addition, they introduced diffusion and gravity terms in their solution to the two-phase transport problem and found consistent results.

Continuing the PINN implications to the Buckley-Leverett problem, Almajid and Abu-Alsaud (2020) also utilize the same methodology observed by Raissi et al. (2017, 2019), Raissi and Karniadakis (2018) for the solution. They fixed their initial, boundary, and collocation points to 160, 80, and 1200 respectively in four cases. Their cases are based on diffusion parameters, observed data, and multiphase trainable parameters (if the model changes it during iteration for better accuracy) and are summarized in Table 4. For the introduction of the diffusive term into NN, they considered the study of Fuks and Tchelepi (2020) who investigated the limitations of PIML approaches. They found that the addition of diffusive terms to generate the hyperbolic equation parabolic improves the overall PIML approach performance. It is to be noted that all the cases are run for two mobility ratios: small and large.

For the first case, the results are found to be inconsistent with the exact solutions. In the second case, however, the predictions are better upto the rarefaction section. After that, the results become inconsistent. For case 3 and case 4, the predictions are consistent with

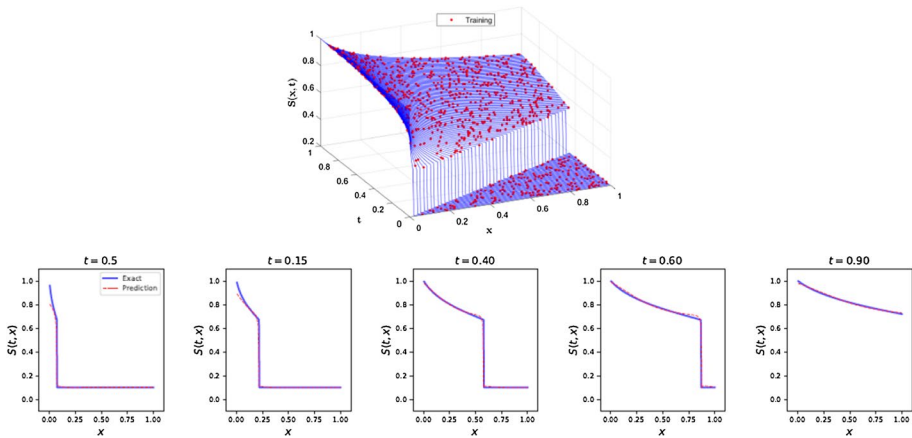


Fig. 8 PINN application on Buckley–Leverett problem with the top image representing PINN solution consistent with the observed data points and bottom figures showing PINN comparison with the exact solution at multiple time realizations—Courtesy: Fraces et al. (2020)

the analytical solutions with case 4 having slight improvements over case 3 because of the introduction of the diffusion terms.

4.1.4 Non-linear diffusivity and Biot's equation

The volumetric flow in porous media under the pressure differential is largely supported by the diffusivity equation which has many applications in hydrocarbon production, ground-water flows, geomechanics, heat mining, etc. (Nick et al. 2013; Bisdorn et al. 2016; Lee et al. 2016; Juanes et al. 2016). The introduction of volumetric fluid with the rock deformation affects the permeability, porosity, and storativity in porous media and is a cause of changes in the flow behavior. This coupling of rock deformation and fluid flow is explained by Biot's poroelasticity equations (Biot 1941; Biot and Willis 1957) which can be solved in a variety of ways from analytical solutions to finite volume discretization, finite difference approximations, and finite element methods (Wang 2017; Masson and Pride 2010; Wenzlau and Müller 2009; Nordbotten 2014; Sokolova et al. 2019; Haga et al. 2012; Choo and Lee 2018; Liu et al. 2018).

Kadeethum et al. (2020), in their work, presented PINN for solving non-linear diffusivity equations and extended their work to solve Biot's equation. The author used the same approach as described earlier with the addition of PDE regularization in the loss function to constrain the training process of the neural network. Also, their work includes both the forward and inverse solution of the defined equations, leveraging the capability of the PINN to solve equations under different conditions and constraints.

The authors also presented the sensitivity of PINN on different hyperparameter choices, and training sizes, and also analyzed the impact of the stochastic variations. For the inverse problem, they also looked at the noise impact. The final results showed that the PINN in the forward modeling problem could predict the pressure and displacement variables in the testing sets for an average loss of $9.05 \times 10^{-5} \pm 3.1 \times 10^{-4}$. For the inverse problem, however, the unknown parameters are predicted with a percentage error of around 1%. As far

Table 4 Summary of the cases

Case	Diffusion parameter	Trainable parameters	Observed data
1	Without	No	Without
2	Without	No	Yes
3	Without	Yes	Yes
4	With	Yes	Yes

as the noise impact is concerned, the PINN is tolerant to a level of 5% (Kadeethum et al. 2020).

4.1.5 Application in medical science

The physics integrating into the neural network has also been investigated in medical science. For example, Sun et al. (2020) applied PINN for the forward problem of idealized blood flow without labeled training data (no simulation data) (Sun et al. 2020). Similarly, the inverse modeling approach of PINN has been utilized in obtaining blood velocity data from the time-resolved concentration data in aneurysms (Raissi et al. 2020; Cai et al. 2021a). In addition integration of physics into the machine learning algorithm has also been used for pressure estimation from 4D flow MRI velocity data (Kissas et al. 2020), and tissue material property prediction (Liu et al. 2020). Similarly, Arzani et al. (2021) utilized PINN for improvements in wall shear stress (WSS) estimation from sparse velocity measurements in diseased arterial flows. The inlet and outlet boundary conditions in this study are considered unknown making the problem mathematically challenging. They evaluated their developed workflow in 1D advection-diffusion transport, 2D and 3D blood flow in aneurysm and stenosis, and determined that the workflow predicts WSS with high accuracies (Arzani et al. 2021). Figure 9 shows the results obtained through the developed PINN workflow in defined scenarios along with the comparison to the sensor data and CFD results. The results show that the PINN predicts accurate solutions with a low amount of data (or no data) provided and are fitting well with the exact solutions and sensor data. The results are fairly smooth with no abruptness between the curves and profiles.

4.1.6 Applications of the PIML approach in different other domains

Similar to the above workflow, Guo et al. (2020) solved the wave equation and kdV–Burgers equations using PINN. The wave equation describes the wave propagation in various disciplines from seismic wave propagation to radio communications (Li et al. 2017; Gu et al. 2018; Kim 2019). In one dimension, the wave equation (Raissi et al. 2018) is defined as:

$$u_{tt} - cu_{xx} = 0, x \in [0, 1], \quad t \in [0, 1] \quad (7)$$

where u is the function differentiated in space x and time t . Also, c represents the velocity term of wave propagation. This equation is bounded by initial and Dirichlet boundary conditions. For the implementation of PINN, the physical constraint function $f(t, x)$ is defined as:

$$f(t, x) : u_{tt} - u_{xx} \quad (8)$$

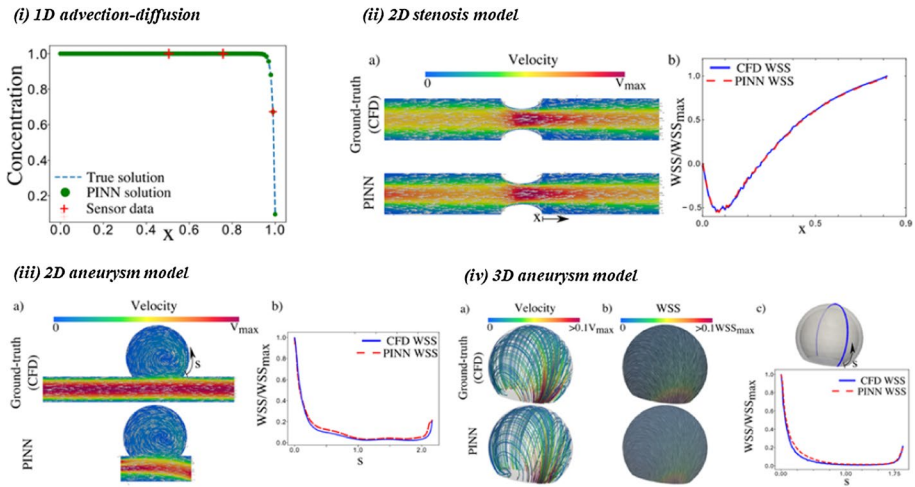


Fig. 9 PINN comparison of velocity and WSS with CFD and sensor data in 1D advection–diffusion transport, 2D and 3D blood flow in aneurysm and stenosis—Courtesy: Arzani et al. (2021)

The training data for PINN is obtained through the initial and boundary conditions, and random Spatio-temporal data. The loss function, i.e. MSE for both the neural network and physical constraint, is minimized to obtain the optimal results. The application of PINN results in a consistent solution of the wave equation with the exact solution as shown in Fig. 10. With this, another researcher Voytan and Sen (2020) solved a similar wave equation with PINN effectively.

With a similar approach, the kdV–Burgers equation is also solved through PINN (Guo et al. 2020). The results were found to be consistent with the exact solution (refer to Fig. 11). The kdV–Burgers equation has many applications relating to liquid flow in elastic tubes containing bubbles, etc. (Samokhin 2017; Ahmad et al. 2020).

Continuing with similar approaches, Kumar et al. (2020) utilized PINN for the profiling of subsurface velocity. Their training dataset includes the data obtained through PySIT and real subsurface receiver data from the Volve field. The PINN in their approach resulted in an MSE of 0.00266, for real data cases, and the predicted values of PINN were found to be matching with the true values (refer to Fig. 12). It is to be noted that the training is performed on limited data i.e. 7.5% of the total dataset.

In another application, Waheed et al. (2021) used PINN for the solution of the eikonal equation. The eikonal equation has a variety of applications in image processing, and seismology (Adalsteinsson and Sethian 1995; Alvino et al. 2007; Malladi and Sethian 1996; Rouy and Tourin 1992; Guo et al. 2019; Grechka et al. 2015; Lagaris et al. 1998). In their approach, the authors trained the network while predicting travel times. Their approach to solving this equation results in an accurate prediction matching the exact solutions. For improving the convergence rates and solution accuracy, the authors used adaptive weighting and adaptive activation functions.

Furthermore, Mishra and Molinaro (2021) used PINN to solve a high-dimensional radiative-transfer problem i.e. the radiative intensity. They highlighted the issue of the curse of dimensionality for the numerical methods and incorporate PINN as a solver in addition to the inclusion of quadrature rules and Sobol low-discrepancy sequences as training points to tackle the curse of dimensionality. They tested the proposed methodology from a simple

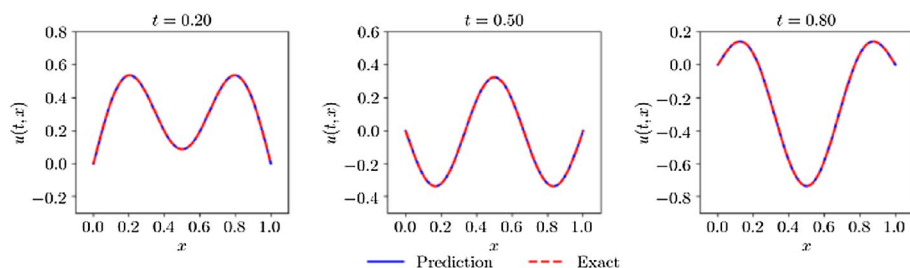


Fig. 10 PINN prediction of the wave equation and comparison of exact solution at different time realization – Courtesy: Guo et al. (2020)

1D monochromatic stationary radiative transfer to a 3D time-dependent polychromatic radiative transfer. Their results show that the proposed method can predict fairly accurate and comparable results with the published results having smaller errors in a low number of epochs (training time). In addition to the above forward problem, they also solved the inverse radiative transfer problem to determine the absorption coefficient from incident radiation measurements with an error of less than 3% (Mishra and Molinaro 2021).

Similarly, the application of PIML workflow has also been extended to the MD simulation in understanding atomistic and molecular-level mechanisms. For example, Islam et al. (2021) used a multi-fidelity physics-informed neural network (MPINN) for the prediction of system properties using MD results i.e. the diffusion coefficients, energy per atom, and system pressure. They also studied an argon-copper nanofluid case for the prediction of viscosity with the variation of volume fraction and temperature. The results showed that the proposed method accurately predicted viscosity with a smaller number of MD simulation runs (refer to Fig. 13) (Islam et al. 2021). Furthermore, Razakh et al. developed a software PND on the basic physics-informed neural network (PINN) which allows the implementation of atomic motion equations, their boundary conditions, and conservation laws in the loss function while training the network. It also has an MD engine for optimizing the performance of a trained network for molecular application (Razakh et al. 2021).

4.2 Open-source libraries for PIML

There are multiple software/libraries available dealing with PINN problems including both the forward and inverse problems. Currently, Python is the commonly used programming language for the majority of PINN problems. Different researchers embedded different libraries in Python acting as solvers and wrappers. Also, these libraries support the platform of Tensorflow and PyTorch which aids in developing an efficient neural network for physical problems. The major libraries include Nvidia Modulus (previously SimNet) (Hennigh et al. 2021), DeepXDE (Lu et al. 2021a) for solving PDE, IDRLnet (Peng et al. 2021)—framework inspired by Nvidia Simnet, PyDENs (Koryagin et al. 2019), NeuroDiffEq (Chen et al. 2020), Gpytorch for gaussian process regression (Gardner et al. 2018), SciANN package (Haghighat and Juanes 2021), TensorDiffEq (McClenny et al. 2021), Elvet (Araz et al. 2021) etc. With this, some of the libraries including ADCME (Xu and Darve 2020) and NeuralPDE (Zubov et al. 2021) are also developed in the Julia framework. The details of the libraries are provided in Table 5.

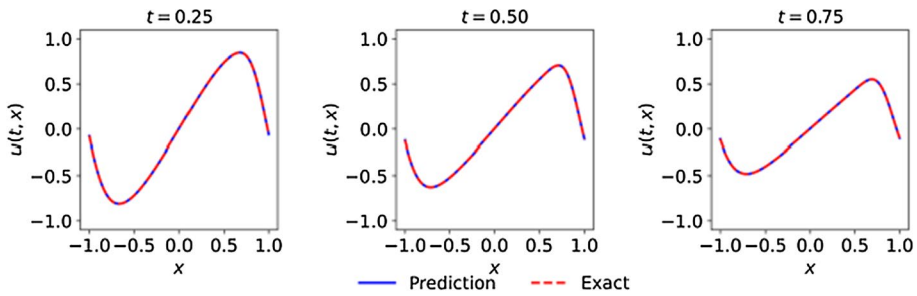


Fig. 11 PINN prediction of kdV–Burgers equation and comparison of exact solution at different time realization—Courtesy: Guo et al. (2020)

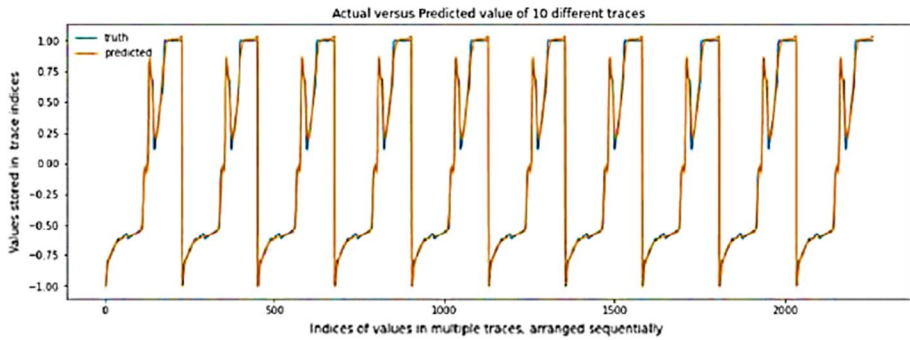


Fig. 12 PINN application for subsurface velocity profiling for 10 different tracers—Courtesy: Kumar et al. (2020)

5 PIML approach advantages, limitations, and recommendations for further development

The development of technology especially in the case of machine learning has paved the way for solving many high-fidelity problems with ease. However, the major hindrance involve in all those techniques is the unavailability of physics and hence the algorithms determine the feature relationships as a black box. With big data, the approach of using data-driven modeling (which in many cases is related to physics-guided learning) is used to understand the underlying structures. Although the recent works determine the success of achieving accurate results with this approach, it is still unclear if the models developed on this platform withstand the generalization of the physical phenomena. Also, big data is not always available for all the problems involved. This presents another restriction in using conventional machine learning algorithms.

To tackle the challenges of model generalization and lack of physics in machine learning algorithms, many researchers work on integrating the PDE with the trained machine learning modules to determine physically consistent and generalized results. These machine learning networks are achieved by regularizing the loss function with the PDE residual and hence constraining the training process. This approach is found to be highly successful in problems dealing with limited data. The PIML approach, in general, offers many advantages including:

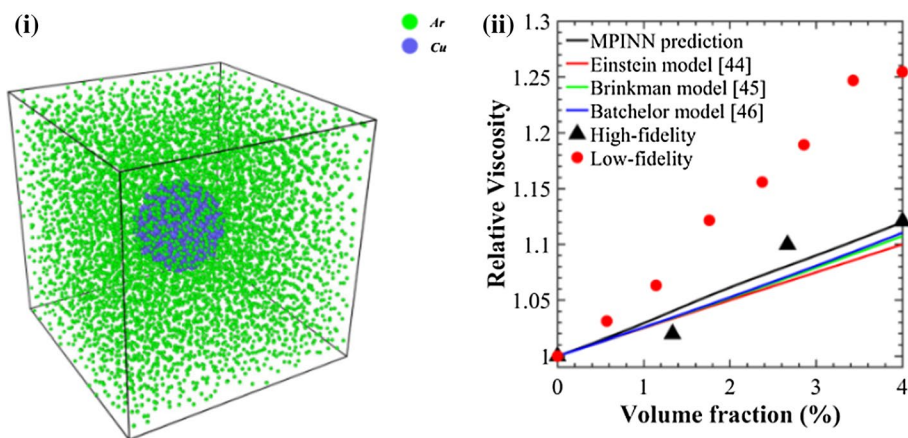


Fig. 13 Ar–Cu nanofluid case (i) initial configuration (ii) MPINN prediction comparisons with theoretical models and multi-fidelity data—Courtesy: Islam et al. (2021)

- Providing a mesh-free approach for solving partial differential equations, hence, reducing the computational efforts and allowing modeling of the complexity of the system. Since PDEs are used in all the science and engineering disciplines, hence, the PINN methodology is well suited for all these forward and inverse modeling problems.
- The PIML allows the spatio-temporal solutions to be completed in a single time step, hence, reducing the efforts which are conventionally done by numerical simulations that usually work on multiple time-stepping.
- In the PIML framework, the PINN, to be specific, can handle and scaled up to very-high dimensional PDEs without incurring the curse of dimensionality. Due to the nature of neural networks, with the increase in dimensionality, the computational time for the solution of the equation within the loss function does not increase exponentially giving it an edge over high-dimensional PDE solutions to numerical solvers.
- PIML works with a limited amount of data. Hence, in comparison with data-driven modeling which requires a large amount of data to learn the physical system, the PINN uses PDE and limited training data to achieve more consistent results.
- The PIML provides a more generalized model with its ability to incorporate physics in the computational processes as compared to data-driven approaches.

However, currently, there are some limitations pertaining to this approach that is recommended to be addressed for future PIML workflow developments:

- Most of the literature on integrating physical laws into the machine learning algorithm is based on solving 1D and 2D problems (and limited 3D problems) with low-dimensional data and less complicated problems. Hence, the applicability of the PIML for high-dimensional data with the more heterogeneous problem is still concerning. The future application of PIML on such high-dimensional structures will prove its further effectiveness.
- In addition to its limited applications in solving PDEs, another important aspect that is challenging is the ability of the PIML approach to consider a strongly coupled PDE system.

Table 5 Major libraries and their description for implementing physics-informed machine learning

Library name	Program- ming language	Backend	Features	Link
Nvidia Modulus	Python	PyTorch	Offers forward and inverse solutions to PDEs both in classical (strong) and weak forms In modulus, the losses are taken as the integrals instead of summation Integral PDEs can also be solved Supports tessellated geometry (TG) and constructive solid geometry (CSG) Supports a variety of architectures including FFNN, Fourier networks, DGM Architecture, Sinusoidal Representation Networks (SiReNs), DeepONet, Super Resolution Net, Pix2Pix Net, and Physics Informed Neural Operator Also offers adaptive activation functions, multiple sampling strategies, Sobolev (gradient enhanced) training, Imposition of exact boundary conditions, learning rate annealing, loss balancing, etc	https://developer.nvidia.com/modulus
DeepXDE	Python	TensorFlow	Solves ordinary/partial/integro/ fractional/stochastic differential equation (ODEs/PDEs/IDEs/IPDEs/sPDEs)	https://github.com/lululxvi/deepxde
		PyTorch	Solves both forward and inverse problems and allows uncertainty quantification	
		JAX PaddlePaddle	PINN with hard constraints and multiscale fourier features are also included Gradient-enhanced PINN (gPINN) and residual-based adaptive sampling are also included Supports different sampling methods such as Latin hypercube, pseudorandom, uniform, Sobol sequence, Halton sequence, and Hammersley sequence Also supports other architectures including DeepONet, DeepM&Mnet, and multi-fidelity neural network (MFNN) It allows solutions under complex geometry with constructive solid geometry (CSG) Five boundary conditions are supported i.e. General BC on any domain, Neumann, Dirichlet, periodic, and Robin	

Table 5 (continued)

Library name	Program- ming language	Backend	Features	Link
IDRLnet	Python	PyTorch	It can solve both forward and inverse PDEs even from noisy measurements Supports the solution of integral differential equations and variational minimization problems Offers adaptive resampling Also supports complex domain geometries Allows samplings in the interior and at the boundary with specified conditions	https://github.com/drl-lab/idrlnet
PyDENs	Python	PyTorch	Offers different network architectures such as DenseNet and Resnet along with the flexibility of building complex networks with fully-connected layers and convolution layers Solves PDEs including common PDEs such as wave; heat; poisson equation, PDEs with trainable coefficients, and parametric families of PDEs The ICs/BCs are enforced in the network similar to PCNN	https://github.com/analysiscenter/pydens
NeuroDiffEq	Python	PyTorch	Sampling can be performed over initial and boundary domains Offers the solutions of ODEs and PDEs up to two-dimensions The ICs/BCs are enforced by construction similar to PCNN Supports solutions in the irregular domain Allows transfer learning flexibility Can control sampling distributions	https://github.com/NeuroDiffGym/neurodiffeq
TensorDiffEq	Python	TensorFlow	Solves both the forward and inverse PDEs problems Offers scalable domains and multi-GPU support Supports self-adaptive collocation method and solvers for fast PINN training Offers interface for defining IC/BCs, strong PDEs, and domains	https://github.com/tensordiffeq/TensorDiffEq

Table 5 (continued)

Library name	Program- ming language	Backend	Features	Link
SciANN	Python	Keras TensorFlow	Supports model fitting, ODEs/PDEs solutions (strong and weak form), and parameter identification (model inversion) Solves variational PDEs Offers adaptive weights and learning rates Supports complex deep neural networks such as FFNN, RNN, and LSTM Can solve problems through transfer learning Offers access to solved applications through SciANN	https://github.com/sciann
Elvet	Python	TensorFlow	Solves ODEs and PDEs including coupled equations with different boundary conditions Supports the solution of variational problems such as functional minimization	https://gitlab.com/elvet/elvet
NeuralPDE.jl	Julia	Julia	Can fit multidimensional data points Solves ODEs, PDEs, IDEs and sPDEs Supports neural adapters and adaptive loss functions Offers quadrature training strategies	https://github.com/SciML/NeuralPDE.jl
ADCME.jl	Julia	TensorFlow Julia	Supports the mixing of the physics-informed loss function with various operators i.e. DeepONets, Graph Neural Operators, Fourier Neural Operators, etc Offers solution of PDEs through neural networks Supports inverse modeling Supports PDEs solutions through the numerical scheme Offers flexibility of parameter identification in stochastic process	https://github.com/kailaix/ADCME.jl

- The stiffness in PDEs is another area that needs improvements.
- Although PIML requires limited data, the data should have high quality that fully describes the IC/BCs of the system. In addition, the need for appropriate equations for IC/BCs puts additional effort into training. Due to this increase in additional boundary data or associated equations, the training time is significantly increased resulting in slow convergence.
- The PIML training involves a completed loss function with multiple variables, hence, presenting a non-convex problem. Hence, the global optimization in such a scenario is still concerning.
- Since PIML optimization is a non-convex problem, therefore, it is difficult to obtain unique solutions when compared with traditional numerical techniques. Therefore, the ML algorithm should be optimized and tuned first to converge to as low loss as possible.
- Compared with numerical solvers for specific real-time multidimensional challenging PDEs including transient, chaotic, and stiff behavior, PIML fails to learn such complex physical challenges. Although transfer learning is helping PIML to predict more accurately in complex real-time conditions, the optimization and updates in the ML structure are necessary to tackle such limitations.
- Challenges of information propagation from initial timesteps to future timesteps (Temporal causality) in a chaotic and transient system possess a challenge in PINN. Although, causal training and sequence learning in the machine learning algorithm are proposed to tackle this issue. However, this area needs further investigation.
- The high computational time required is still a concern, although, various approaches such as the use of adaptive functions or training on neural operators, etc., have been presented to minimize the computational time, yet, this area needs further improvements.
- There is a trade-off between algorithm complexity and accuracy. Finer grids and high resolution may reduce the truncation errors, however, sometimes the optimization becomes unstable.

Since the PIML methodology is developing, therefore, it is expected that the current limitations will be addressed in the coming years, and a more stable and generalized framework will be developed for multi-fidelity problems. In general, the recommendation is to test different combinations of optimizers, sampling type, sampling data, appropriate weight distributions to multiple loss terms, and types of PDE based on the defined problem to achieve an accurate and physically consistent result with reduced computational efforts. Furthermore, coupling PIML with traditional solvers can be another area of investigation for increased prediction capabilities. In addition, the research and applications on other machine learning algorithms (currently FFNN is extensively used and researched) must be expanded to achieve more generalized solutions with better accuracy and reduced computational efforts.

Considering all this discussion, one question arises which is what type of methodology should be used for tackling the science and engineering problems. The answer most probably lies in the amount of data and the resources available. Considering the high amount of data and its availability, it is important to consider the data-driven solutions in high-dimensional problems with complex systems involved, since, the applicability of PIML on complex systems has not been studied in depth for different physical processes. For limited data, with low-dimensional features and multiphysics, the PIML should be the first choice of implementation. A hybrid approach should also be considered, that is implementing the

data-driven modeling with the PIML methodology which may control the high dimensional problem of the PIML implementation. Hence, more research is needed to be done.

6 Conclusions

This paper reviews the application of physics into machine learning modules for efficient training of the system. In this paper, different methodologies with their respective examples are provided to achieve the physics in machine learning, especially the neural network model. The paper also presents the applications, advantages, and limitations of PIML approaches to various science and engineering problems. Considering the discussion presented in this review, the following major conclusions can be drawn:

- Data and physics play a vital role in modeling any physical system. Different techniques such as analytical, numerical, and machine learning are usually applied to solve the underlying problems. Machine learning, in particular, allowed to solve data-physics problems with high certainty, accuracy, and reduced computational efforts.
- The amount and quality of data help in decision making a machine learning process. Big data with limited physics can be solved through a data-driven modeling approach. While, implementing the physics, as in the case of limited data, through the PIML approach is proved to be an effective approach in dealing with uncertain and complex problems.
- The PIML workflow solves the problem by minimizing the loss of PDE and trained networks obtained through observed/simulated data in the system.
- This PIML workflow is found to be consistent with the exact solutions of many PDEs in both the forward and inverse modeling approach showing the capability of the workflow in dealing with various science and engineering problems.
- The PIML architecture depends upon many factors including the size of the training data, the quality of data, additional loss terms, updated physics, stochastic variations, hyperparameters choices, and noise in the system.
- Most of the literature shows the application of PIML to relatively simple problems with the low-dimensional dataset.

With such advancements in machine learning technologies, it is expected that the limitations of the proposed PIML will be studied in depth in future studies and a general framework leveraging the capabilities of both the PIML and data-driven complex solutions will be put forward. For now, PIML offers high advantages of integrating physics into machine learning training, opening avenues for future research and problem optimizations.

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